

ASSIGNMENT - 3

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(1) Determine the multiplicative inverse mod 26 for each of the following integers

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25.

$$3^{-1}$$

$$26 = 3(8) + 2 \Rightarrow 26 + 3(-8) = 2 \rightarrow \textcircled{1}$$

$$3 = 2(1) + 1 \Rightarrow 3 + 2(-1) = 1 \rightarrow \textcircled{2}$$

Substitute $\textcircled{2}$ in $\textcircled{1}$

$$3 + (3(-8) + 26)(-1) = 1$$

$$3 + 3(8) + 26(-1) = 1$$

$$3(9) + 26(-1) = 1$$

$$\boxed{3^{-1} = 9}$$

$$5^{-1}$$

$$26 = 5(5) + 1$$

$$26 + 5(-5) = 1$$

$$26 - 5 = 21$$

$$\boxed{5^{-1} = 21}$$

$$7^{-1}$$

$$26 = 7(3) + 5$$

$$5 = 26 + 7(-3)$$

$$7 = 5(1) + 2$$

$$2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$1 = 5 + 2(-2)$$

$$5 + 2(-2) = 1$$

$$5 + [7 + 5(-1)](-2) = 1$$

$$5 + 7(-2) + 5(2) = 1$$

$$5(3) + 7(-2) = 1$$

$$[26 + 7(-3)]3 + 7(-2) = 1$$

$$26(3) + 7(-9) + 7(-2) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15 \quad \boxed{7^{-1} = 15}$$

9^{-1}

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + 8(26 + 9(-2))(-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$\boxed{9^{-1} = 3}$$

11^{-1}

$$26 = 11(2) + 4$$

$$26 + 11(-2) = 4 \rightarrow \textcircled{1}$$

$$11 = 4(2) + 3$$

$$11 + 4(-2) = 3 \rightarrow \textcircled{2}$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$4 + [11 + 4(-2)](-1) = 1$$

$$4 + 11(-1) + 4(2) = 1$$

$$4(3) + 11(-1) = 1$$

$$(26 + 11(-2))3 + 11(-1) = 1$$

$$26(3) + 11(-6) + 11(-1) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 1$$

$$\#2019$$

$$\boxed{11^{-1} = 19}$$

15^{-1}

$$26 = 15(1) + 11$$

$$26 + 15(-1) = 11$$

$$15 = 11(1) + 4$$

$$15 + 11(-1) = 4$$

$$11 = 4(2) + 3$$

$$11 + 4(-2) = 3$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$15 + 11(-1) + [11 + 4(-2)](1) = 1$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + [26 + 15(-1)](-2) + [15 + 11(-1)](2) = 1$$

$$15 + 26(-2) + 15(2) + 11(-2) = 1$$

$$15(5) + 26(-2) + [26 + 15(-1)](-2) = 1$$

$$15(5) + 26(-2) + 26(-2) + 15(2) = 1$$

$$15(7) + 26(-4) = 1$$

$$\boxed{15^{-1} = 7}$$

 17^{-1}

$$26 = 17(1) + 9$$

$$\Rightarrow 26 + 17(-1) = 9$$

$$17 = 9(1) + 8$$

$$\Rightarrow 17 + 9(-1) = 8$$

$$9 = 8(1) + 1$$

$$\Rightarrow 9 + 8(-1) = 1$$

$$26 + 17(-1) + [17 + 9(-1)](-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-2) + (26 + 17(-1))(1) = 1$$

$$26 + 17(-2) + 26(1) + 17(1) = 1$$

$$26(2) + 17(-3) = 1 \Rightarrow 26 - 3 = 23$$

$$\boxed{17^{-1} = 23}$$

2. Choose any string of your choice to encrypt using the affine cipher with key values, multiplier 5 & additive 8.

Plaintext: GOOD LUCK.

For Encryption.

$$C = [k_1 * P + k_2] \bmod 26$$

Given $k_1 = 5$, $k_2 = 8$.

Plain text:

	$k_1 \times P$	$+ k_2$		m
G 6	$5 \times 6 = 30$	$30 + 8$	38	a
O 14	$5 \times 14 = 70$	$70 + 8$	78	a
O 14	$5 \times 14 = 70$	$70 + 8$	78	x
D 3	$5 \times 3 = 15$	$15 + 8$	23	L
L 11	$5 \times 11 = 55$	$55 + 8$	63	e
U 20	$5 \times 20 = 100$	$100 + 8$	108	y
C 2	$5 \times 2 = 10$	$10 + 8$	18	
K 10	$5 \times 10 = 50$	$50 + 8$	58	

Ciphertext = MAA x L E S y

$$C = (k_1 * P + k_2) \bmod 26$$

$$P = \frac{C - k_2}{k_1} \bmod 26$$

$$P = (C - k_2) \cdot k_1^{-1} \bmod 26$$

$$P = (C - k_2) \cdot k_1^{-1} \bmod 26$$

$$P = (C - k_2) \cdot k_1^{-1} \bmod 26$$

$$P + (1)P = 26$$

$$8 + (1)P = 26$$

$$1 + (1)P = 26$$

$$1 = (1)P + (1)P + (1)P + 26$$

$$1 = (1)P + (1)P + (1)P + 26$$

$$1 = (1)P + (1)P + (1)P + 26$$

$$E = F$$

3. Encrypt "HT" using the affine cipher with
 $a = 11$ & $b = 5$.

For Encryption

$$C = K (k_1 * P + k_2) \bmod 26$$

$$k_1 = 11 \quad k_2 = 5$$

P.T

		$k_1 * P$	$+ k_2$	$\bmod 26$	Cipher
21	7	$11 * 7 = 77$	$77 + 5 = 82$	4	E
1	8	$11 * 8 = 88$	$88 + 5 = 93$	15	P

$$\text{Cipher text} = KE P.$$

(4) The ciphertext UCR was encrypted using the affine function $C = 9P + 2 \bmod 26$.
 Find the plaintext $P = (8)8 + (1)25 + 2$

=> For Decryption

$$P = K^{-1} (C - k_2) \bmod 26$$

$$k_1 = 9 \quad k_2 = 2$$

$$9^{-1} \bmod 26$$

$$26 = 9(2) + 8 \Rightarrow 26 + 9(-2) = 8$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + (26 + 9(-2))(-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) \equiv 26(1) \\ 9^{-1} \bmod 26 = 3$$

Ciphertext		$K^{-1} (C - k_2) \bmod 26$	P
U	20	$3(20 - 2) = 54$	2
C	2	$3(2 - 2) = 0$	0
R	17	$3(17 - 2) = 45$	19

$$PT = CAT$$

5. Decrypt the cipher text using Affine cipher
 K D Y X J U W D E X Z Y using a key k_1 (mul) &
 additive

$$\rightarrow P = K^{-1} [C - K_2] \text{ mod } 26.$$

plaintext: K D Y X J U W D E X Z Y

$$k_1 = 3 \quad k_2 = 4$$

$$3^{-1} \text{ mod } 26.$$

$$26 = 3(8) + 2 \Rightarrow 26 + 3(-8) = 2$$

$$3 = (2)(1) + 1$$

$$3 + (-1)2 = 1$$

$$3 + (26 + 3(-8)) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$3(9) = 26(1)$$

$$\boxed{3^{-1} \text{ mod } 26 = 9}$$

cr		$K_1^{-1}(C - K_2)$	mod 26	P
K	10	$9(10 - 4) = 54$	2	P
D	3	$9(3 - 4) = -9$	17	R
Y	24	$9(24 - 4) = 180$	24	Y
X	23	$9(23 - 4) = 171$	15	P
J	9	$9(9 - 4) = 45$	19	T
U	20	$9(20 - 4) = 144$	14	O
W	22	$9(22 - 4) = 162$	6	G
D	3	$9(3 - 4) = -9$	17	R
E	4	$9(4 - 4) = 0$	0	A
X	23	$9(23 - 4) = 171$	15	P
Z	25	$9(25 - 4) = 189$	7	H
Y	24	$9(24 - 4) = 180$	24	Y

(6) Use $K_1 = 13$, $K_2 = 4$ (Even though it is not awarded key)

try to encrypt)

Encrypt the string 'input' & 'alter' using the what is the issue.

For Encryption:

$$C = E(K_1 * P + K_2) \bmod 26 \quad K_1 = 13, K_2 = 4$$

PT		$K_1 * P$	$+ K_2$	$\bmod 26$	E
I	8	$13 \times 8 = 104$	108	4	E
N	13	$13 \times 13 = 169$	173	17	R
P	15	$13 \times 15 = 195$	199	17	E
U	20	$13 \times 20 = 260$	264	4	R
T	19	$13 \times 19 = 247$	251	17	R
A	0	$13 \times 0 = 0$	4	4	E
L	11	$13 \times 11 = 143$	147	17	R
T	19	$13 \times 19 = 247$	251	17	R
E	4	$13 \times 4 = 52$	56	4	E
R	17	$13 \times 17 = 221$	225	17	R

$K_1 = 13$ is not co-prime with 26.

The entire 26 letter alphabet is compressed into just two ciphertext letters (E & R)

7) Find the Inverse Matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ mod 26

$$\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} = (3 \times 7) - (5 \times 2) = 21 - 10 = 11$$

$$11^{-1} \text{ mod } 26 = 26 = 11(2) + 4$$

$$26 + 11(-2) = 4$$

$$11 = 4(2) + 3$$

$$11 + 4(-2) = 3$$

$$4 = 3(1) + 1$$

$$1 = (26 - 11(2)) - (11) = -6(11) - (11) + 3 \times 26$$

$$1 = (-7) 11 + 3 \times 26 \quad -7 + 26 = 19$$

$$\boxed{11^{-1} \text{ mod } 26 = 19}$$

$$19 \begin{bmatrix} 7 & 2 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 13 & 3 \\ -9 & 5 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8.) Find the inverse of the following matrix. The entries are considered modulo 26. 11 13
2 3

$$\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} = (11 \times 3) - (13 \times 2) = (33 - 26) = 7$$

$$7^{-1} \text{ mod } 26 \Rightarrow 26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$1 = 3(26 - 7(3)) - 7(7)$$

$$1 = (-11)7 + 3(26)$$

$$-11 + 26 = 15$$

$$7^{-1} \text{ mod } 26 = 15$$

$$15 \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 165 & 195 \\ 30 & 45 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

[9] Is the key valid: $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix} = (2 \times 21) - (13 \times 14) = (42 - 182) = -140$$

Determinant is not co-prime with 26.

(16) Encrypt using 2×2 Hill cipher for the plain text "SOLVED" using

$$C = P + K \pmod{26}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = (3 \times 11) - (7 \times 8) = (33 - 56) = -23$$

23 is non-zero

8 & 23 is co-prime with 26

$P \Rightarrow S O L V E D$

$$(1) P = \begin{bmatrix} S \\ O \end{bmatrix} = \begin{bmatrix} 18 \\ 14 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 18 \\ 14 \end{bmatrix} = \begin{bmatrix} 3 \times 18 + 8 \times 14 \\ 7 \times 18 + 11 \times 14 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \pmod{26} = \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{bmatrix} K \\ U \end{bmatrix}$$

$$(2) P = \begin{bmatrix} L \\ V \end{bmatrix} = \begin{bmatrix} 11 \\ 21 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} = \begin{bmatrix} 3 \times 11 + 8 \times 21 \\ 7 \times 11 + 11 \times 21 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 22 \end{bmatrix} = \begin{bmatrix} T \\ W \end{bmatrix}$$

$$(3) P = \begin{bmatrix} E \\ D \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \times 4 + 8 \times 3 \\ 7 \times 4 + 11 \times 3 \end{bmatrix} = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \pmod{26} = \begin{bmatrix} 10 \\ 9 \end{bmatrix} = \begin{bmatrix} K \\ J \end{bmatrix}$$

Ciphertext = KUTWKJ